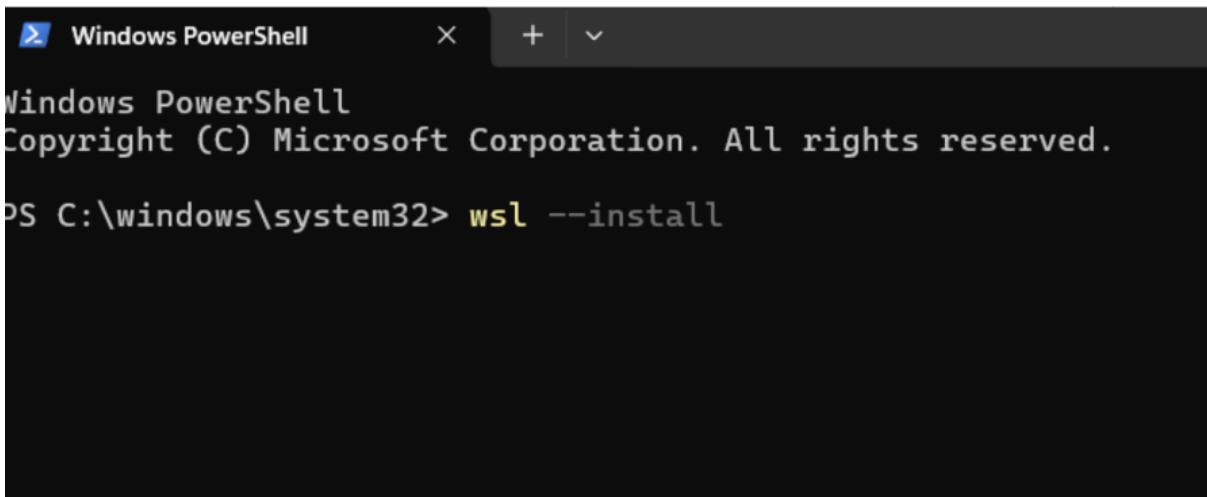
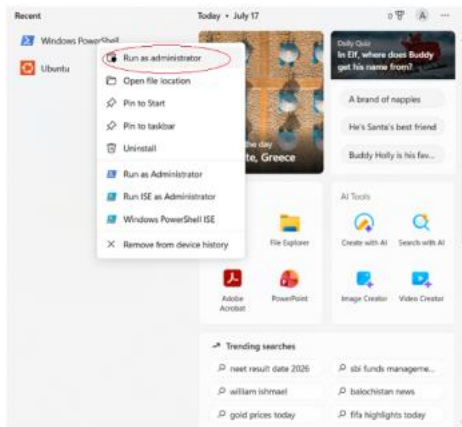
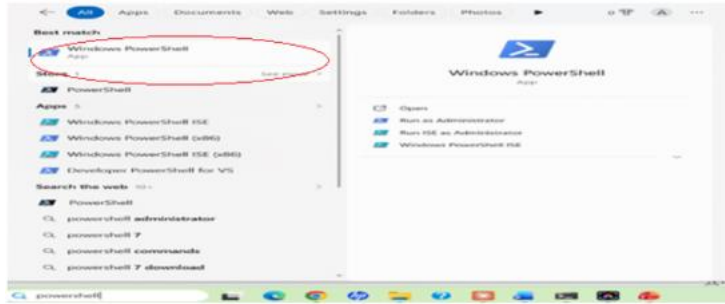


OSSP (Operating Systems And Systems Programming) - 25CS2104E

Skill – 1:

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile



```

hp@Hima:~$ sudo apt update
[sudo] password for hp:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [222 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [772 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [43.3 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [188 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [320 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [61.5 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [336 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [68.5 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [15.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6,472 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [506 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [77.4 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,330 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [239 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [55.0 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.7 kB]
Fetched 4,844 kB in 3s (1,615 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hp@Hima:~$

```

```

hp@Hima:~$ sudo apt list --upgradable
Listing... Done
Building dependency tree... Done
Reading state information... Done
74 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

Packagename	Version	Old Version	Size
linux-headers-generic-hwe-22.04	6.8.0-40.40~22.04.1~22.04.1	6.8.0-40.40~22.04.1~22.04.1	17.1 kB
linux-hwe-6.8-headers-6.8.0-40	6.8.0-40.40~22.04.1	6.8.0-40.40~22.04.1	1,308 kB
linux-image-generic-hwe-22.04	6.8.0-40.40~22.04.1~22.04.1	6.8.0-40.40~22.04.1~22.04.1	71.2 kB
linux-modules-extra-6.8.0-40-generic-hwe-22.04	6.8.0-40.40~22.04.1	6.8.0-40.40~22.04.1	44.7 kB
linux-modules-extra-generic-hwe-22.04	6.8.0-40.40~22.04.1~22.04.1	6.8.0-40.40~22.04.1~22.04.1	36.9 kB
linux-modules-generic-hwe-22.04	6.8.0-40.40~22.04.1~22.04.1	6.8.0-40.40~22.04.1~22.04.1	285 kB
linux-libc-dev	5.15.0-136.146	5.15.0-136.146	44.8 kB
linux-tools-6.8.0-40-generic	6.8.0-40.40~22.04.1	6.8.0-40.40~22.04.1	904 kB
linux-tools-generic-hwe-22.04	6.8.0-40.40~22.04.1~22.04.1	6.8.0-40.40~22.04.1~22.04.1	4,096 B
initramfs-tools	0.140ubuntu13.13	0.140ubuntu13.13	418 kB
libc6	2.35-0ubuntu3.9	2.35-0ubuntu3.8	1,484 kB
libc6-dev	2.35-0ubuntu3.9	2.35-0ubuntu3.8	2,157 kB
libgcc-s1	12.3.0-1ubuntu1-2.4	12.3.0-1ubuntu1-2.4	49.6 kB
libglib2.0-0	2.72.4-0ubuntu2.3	2.72.4-0ubuntu2.3	16.8 kB
libgl1-mesa-glx	2.73.0-1ubuntu2.4	2.73.0-1ubuntu2.4	84.0 kB
libgomp1	3.0.2-0ubuntu3.12	3.0.2-0ubuntu3.12	49.0 kB
openssl	2.49.1-0ubuntu3.12	2.49.1-0ubuntu3.12	1,286 kB
74 packages \$	2.9.4-0ubuntu6.2.11	2.9.1-0ubuntu3.11	1,019 kB

GitHub:

Himavarshini-07 / 2520030141-OSSP

Type to search

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+

+

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

2520030141-OSSP

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

Code

Himavarshini-07 Delete Pratical/1

1065735 · last week

22 Commits

Pratical

Delete Pratical/1

last week

SKILL

Delete SKILL/2520030141-hima 30date7month.pdf

2 weeks ago

README.md

Update README.md

3 weeks ago

Screenshot 2026-08-03 203208.png

Add files via upload

3 weeks ago

README

2520030141-OSSP

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
hp@Hima: ~/SKILL
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        // Execute ls command
        execlp("ls", "ls", "-l", NULL);

        // Runs only if exec fails
        perror("exec failed");
        exit(1);
    }
    else {
        // Parent process
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        // Wait for child
        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
hp@Hima:~/SKILL$ nano Skill1.c
hp@Hima:~/SKILL$ gcc Skill1.c -o Skill1
hp@Hima:~/SKILL$ ./Skill1
Parent Process
Parent PID: 1157
Child Process
Child PID: 1158
Parent PID: 1157
total 40
-rwxr-xr-x 1 hp hp 16184 Aug 22 05:26 SimpleCommandloop
-rw-r--r-- 1 hp hp 386 Aug 22 05:26 SimpleCommandloop.c
-rwxr-xr-x 1 hp hp 16296 Aug 22 05:45 Skill1
-rw-r--r-- 1 hp hp 764 Aug 22 05:44 Skill1.c
Child process completed.
hp@Hima:~/SKILL$
```

